

14.1 Measuring Temperature

Question Paper

Course	CIE A Level Physics
Section	14. Temperature
Topic	14.1 Measuring Temperature
Difficulty	Easy

Time allowed: 50

Score: /34

Percentage: /100

Question 1a

Use the words in the box to complete the gaps in the definition of thermal energy.

higher	possessed	lower	gained	temperature	heat	movement
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Thermal energy is defined as the energy _____ by an object due to its _____. Thermal energy is transferred from a region of _____ temperature to a region of _____ temperature.

[4 marks]**Question 1b**

Identify, by placing a tick (✓) next to the correct statements in Table 1.1 about energy and thermal energy.

Table 1.1

Statements	Place a tick (✓) here if correct
The conservation of energy states that energy is never created or destroyed, only transferred from one form to another.	
When a thermometer is placed in a beaker of boiling water, the thermometer reading increases. This is because the liquid in the thermometer is a lot warmer than the water.	
Temperature tells us the magnitude of energy flow when two regions are in contact.	
The mechanism by which the thermal energy is transferred is by either conduction, convection or radiation.	

[2 marks]

Question 1c

Add an arrow to Fig. 1.1 to indicate the direction of energy flow between the 100 °C and the 0 °C region.

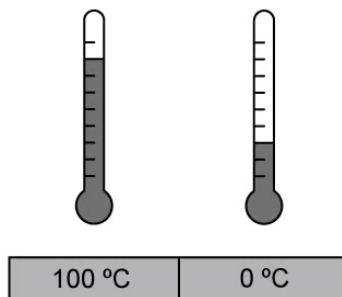


Fig. 1.1

[1 mark]

Question 1d

The sentences in Table 1.2 below describe the process of thermal energy transfer from a hotter region to a lower region

Order the sentences chronologically, by writing numbers 1–4 in the right column, to describe this process.

Table 1.2

Statements about thermal energy transfer	Number the stages of the process
Two regions need to be in contact for thermal energy transfer to occur.	
The final temperature when two regions are in thermal equilibrium depends on the initial temperature difference between them.	
Thermal energy is transferred from the hot region to the cold region.	
The hotter region will cool down and the cooler region will heat up until they reach the same temperature.	

[4 marks]

Question 2a

There are several properties that can be used to measure temperature.

Identify, by placing ticks (✓) in Table 1.1, next to the properties that can be used to measure temperature.

Table 1.1

Property	Place a tick (✓) here
The colour of an object	
The density of a liquid	
The resistance of a metal	
The change in position of a solid	
The volume of a gas at constant pressure	
The e.m.f. of a thermocouple	

[4 marks]

Question 2b

The density of a liquid can be used to measure the temperature of a fluid.

Use the words in the box below to complete the sentences,

density	volume	expands	contracts	measured	counted	length	colour
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A liquid-in-glass thermometer depends on the _____ change of a liquid. It consists of a thin glass capillary tube containing a liquid that _____ with temperature.

A scale along the side of the tube allows the temperature to be _____ based on the _____ of liquid within the tube.

[4 marks]

Question 2c

Temperature can be measured by observing the volume of a gas at a constant pressure.

State the correct relationship for Charles Law that shows this.

[1 mark]

Question 2d

The resistance of a metal in a device such as a thermistor can be used to measure temperature.

Sketch on Fig. 1.1 the correct shape of the resistance–temperature graph for a thermistor.

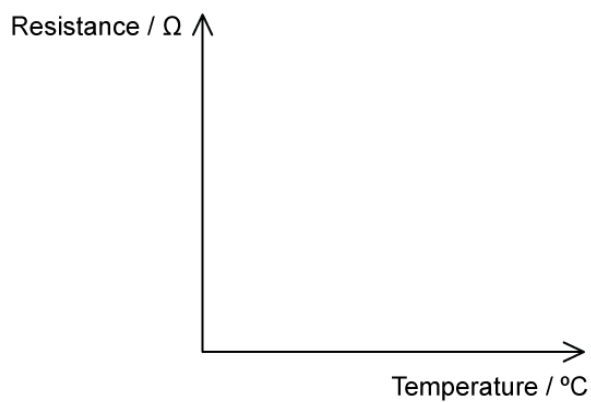


Fig. 1.1

[2 marks]

Question 3a

A thermocouple is an electrical device used as the sensor of a thermometer.

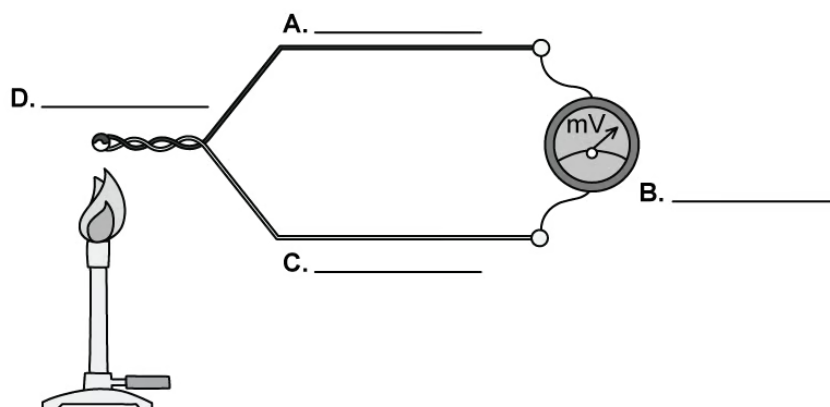


Fig. 1.1

Add the following labels to the correct place on the diagram in Fig. 1.1:

- Voltmeter
- Dissimilar metal 1
- Dissimilar metal 2
- Junction

[4 marks]

Question 3b

The e.m.f. and temperature are not directly proportional to each other in a thermocouple.

Sketch a curve to show this on the axes of Fig. 1.2.

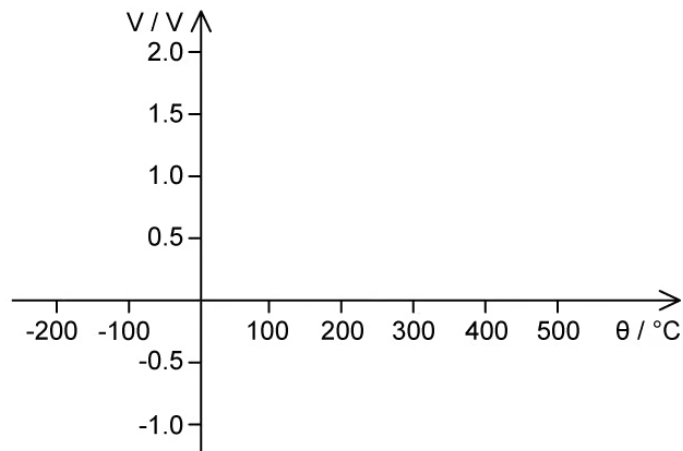


Fig. 1.2

[2 marks]

Question 3c

Define absolute zero.

[2 marks]

Question 3d

Temperature can be measured using both the celsius and Kelvin scales.

(i)

Convert 100°C into Kelvin

[2]

(ii)

Convert -221 K into $^{\circ}\text{C}$.

[2]

[4 marks]

Model Answers: Easy

1a

(a) A correctly completed definition for thermal energy that would score 4 marks should look like this:

- Thermal energy is defined as the energy **possessed**; [1 mark]
- by an object due to its **temperature**; [1 mark]
- Thermal energy is transferred from a region of **higher**; [1 mark]
- temperature to a region of **lower** temperature; [1 mark]

[Total: 4 marks]

1b

(b) A table that would score 2 marks should look like this:

Statements	Place a tick (✓) here if correct
The conservation of energy states that energy is never created or destroyed, only transferred from one form to another.	✓ [1 mark]
When a thermometer is placed in a beaker of boiling water, the thermometer reading increases. This is because the liquid in the thermometer is a lot warmer than the water.	
Temperature tells us the magnitude of energy flow when two regions are in contact.	
The mechanism by which the thermal energy is transferred is by either conduction,	✓ [1 mark]

convection or radiation.

- One mark for each correct statement ticked:
 - The conservation of energy states that energy is never created or destroyed, only transferred from one form to another; [1 mark]
 - The mechanism by which the thermal energy is transferred is by either conduction, convection or radiation; [1 mark]

[Total: 2 marks]

The second statement is incorrect because:

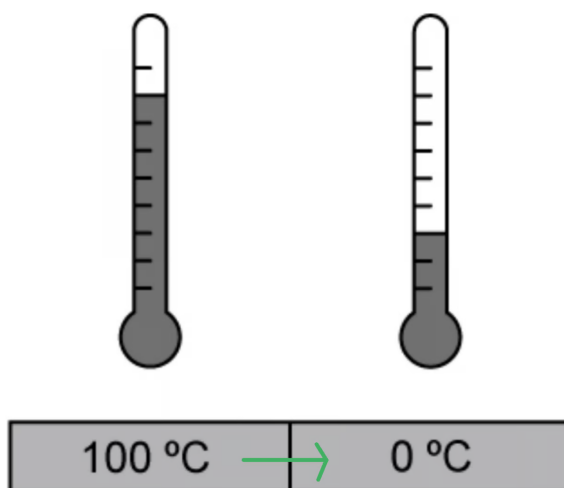
"The liquid in the thermometer is a lot **cooler (and not warmer)** than the water"

The third statement is incorrect because:

"Temperature tells us the **direction (and not magnitude)** of energy flow when two regions are in contact."

1c

(c) A diagram that shows the arrow in the correct direction and would score 1 mark should look like this:



- Arrow drawn from left to right, from 100 °C to 0 °C; [1 mark]

[Total: 1 mark]

1d

(d) A table with the sentences numbered in the correct order that would score 4 marks should look like this:

Statements about thermal energy transfer	Number the stages of the process
Two regions need to be in contact for thermal energy transfer to occur.	1
The final temperature when two regions are in thermal equilibrium depends on the initial temperature difference between them.	4
Thermal energy is transferred from the hot region to the cold region.	2
The hotter region will cool down and the cooler region will heat up until they reach the same temperature.	3

- One mark for each statement in the table numbered correctly up to total four marks

OR

- One mark for placing:
 - 'Two regions need to be in contact for thermal energy transfer to occur' before 'Thermal energy is transferred from the hot region to the cold region'; [1 mark]
- One mark for placing:
 - 'Thermal energy is transferred from the hot region to the cold region' before 'The hotter region will cool down and the cooler region will heat up until they reach the same temperature'; [1 mark]
- One mark for placing:

- o 'The hotter region will cool down and the cooler region will heat up until they reach the same temperature' before 'The final temperature when two regions are in thermal equilibrium depends on the initial temperature difference between them'; [1 mark]

[Total: 4 marks]

2a

(a) A table with tick marks in the correct place that would score 4 marks should look like this:

Property	Place a tick (✓) here
The colour of an object	
The density of a liquid	✓ [1 mark]
The resistance of a metal	✓ [1 mark]
The change in position of a solid	
The volume of a gas at constant pressure	✓ [1 mark]
The e.m.f. of a thermocouple	✓ [1 mark]

- One mark for each correct property ticked

[Total: 4 marks]

2b

(b) A complete set of sentences describing how the density of a liquid can be used to measure its temperature that would score 4 marks should look like this:

- A liquid-in-glass thermometer depends on the **density** change of a liquid; [1 mark]
- It consists of a thin glass capillary tube containing a liquid that **expands** with temperature; [1 mark]
- A scale along the side of the tube allows the temperature to be **measured**; [1 mark]

[Total: 4 marks]

2a

(a) A table with tick marks in the correct place that would score 4 marks should look like this:

Property	Place a tick (✓) here
The colour of an object	
The density of a liquid	✓ [1 mark]
The resistance of a metal	✓ [1 mark]
The change in position of a solid	
The volume of a gas at constant pressure	✓ [1 mark]
The e.m.f. of a thermocouple	✓ [1 mark]

- One mark for each correct property ticked

[Total: 4 marks]

2b

(b) A complete set of sentences describing how the density of a liquid can be used to measure its temperature that would score 4 marks should look like this:

- A liquid-in-glass thermometer depends on the **density** change of a liquid; [1 mark]
- It consists of a thin glass capillary tube containing a liquid that **expands** with temperature; [1 mark]
- A scale along the side of the tube allows the temperature to be **measured**; [1 mark]
- based on the **length** of liquid within the tube; [1 mark]
- One mark for each word in the correct place

[Total: 4 marks]

2c

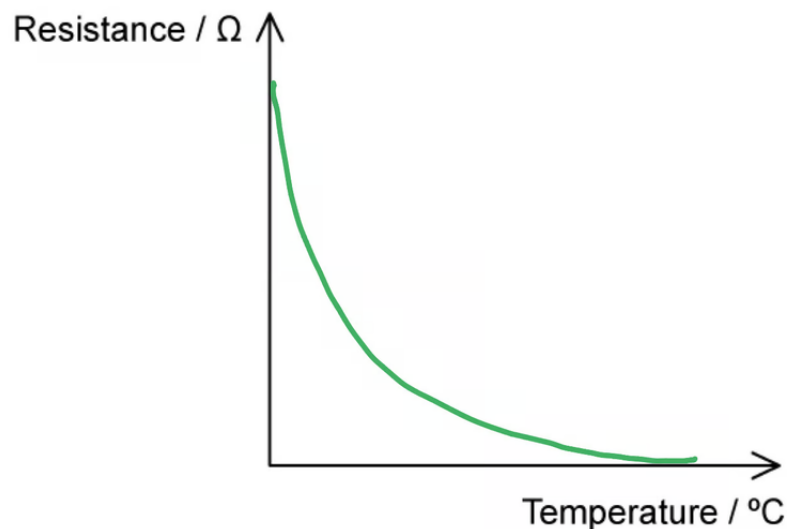
(c) The correct relationship for Charles' Law to show the volume of a gas at constant pressure is:

- Volume / $V \propto$ Temperature / T ; [1 mark]

[Total: 1 mark]

2d

(d) A correct resistance-temperature graph that would score 2 marks should look like this:

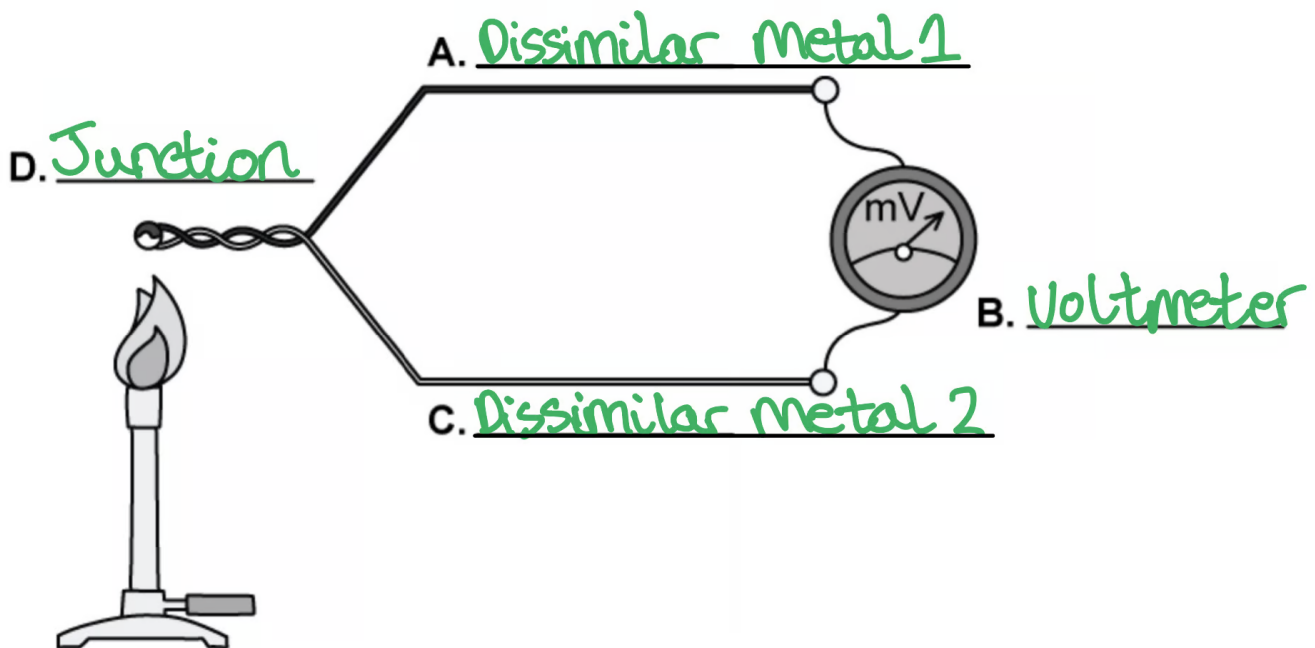


- A curve / a non-linear graph is drawn on the axes; [1 mark]
- The curve has a high resistance for a zero temperature **AND** a low resistance for a high temperature; [1 mark]

[Total: 2 marks]

3a

(a) A correctly labelled diagram of a thermocouple that should score 4 marks would look like this:

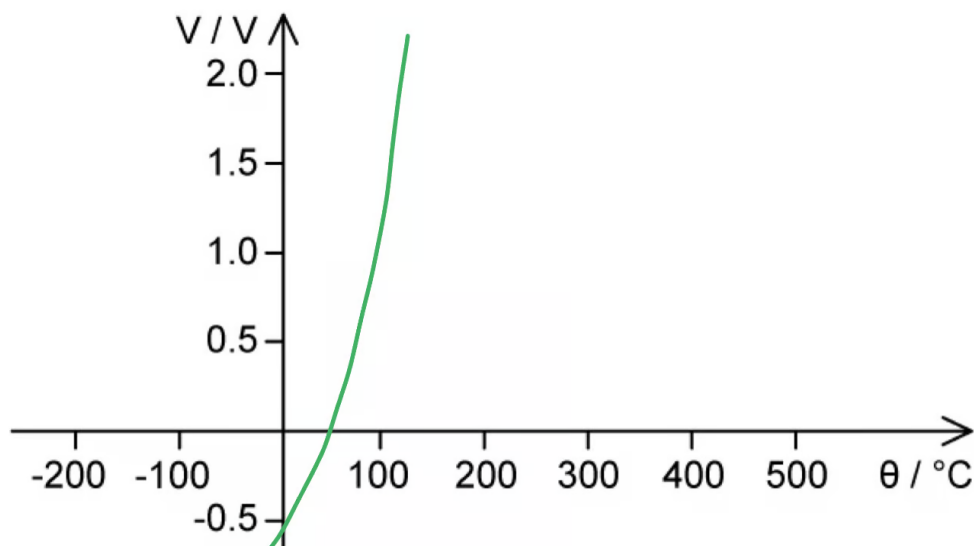


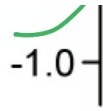
- One mark for each correct label A, B, C and D

[Total: 4 marks]

3b

(b) A sketch on the axes to show the relationship between e.m.f. and temperature of a thermocouple that would score 2 marks should look like this:





- The line drawn is a curve; [1 mark]
- The curve goes from bottom left to top right; [1 mark]

[Total: 2 marks]

3c

(c) The definition of absolute zero is:

- The lowest temperature possible; [1 mark]
- Equal to 0 K or -273.15°C ; [1 mark]

[Total: 2 marks]

3d

(d)

(i) Convert 100°C into Kelvin:

- Temperature in K = temperature in $^{\circ}\text{C} + 273.25$ [1 mark]
- Temperature in K = $100 + 273.15$
- So, $100^{\circ}\text{C} = 373.15\text{ K}$ [1 mark]

(ii) Convert -221 K into $^{\circ}\text{C}$:

- Temperature in $^{\circ}\text{C} = \text{temperature in K} - 273.15$ [1 mark]
- Temperature in $^{\circ}\text{C} = -221 - 273.15$
- So, $-221\text{ K} = -494.15^{\circ}\text{C}$ [1 mark]

[Total: 4 marks]

